Ádám Fodor

PhD Research Scientist

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Class B driving license

Budapest, Hungary

About Me

PhD candidate with over 11 years of research and development experience in machine learning and computer vision. Focused on Human-Centered AI for personality research and affective computing. Experienced lecturer and supervisor.

Languages

- Hungarian – native
- English – fluent

Skills

Programming & Development
- Python, Bash scripting
- DevOps (Git, GitHub Actions, Docker, Singularity)

Machine Learning & Deep Learning
- Multimodal DL (visual, acoustic, textual)
- MLOps (PyTorch, Lightning, MLflow, W&B, HF Spaces)
- Transformers (LLMs, VLMs, Ollama, RAG systems)

Cloud & Compute & Deployment
- Cloud platforms: IBM Cloud, AWS S3
- RESTful APIs, Flask, ONNX, distributed training
- Data and feature engineering, visualization

Soft skills
- Experimental design & hypothesis testing
- Technical writing & scientific communication

Awards

- 1st place Students' Scientific Association Conference (TDK), 2018
- Special award, Morgan Stanley, 2018
- Student TDK supervision award, 2023

Experience

Research Scientist — since 09/2017

Neural Information Processing Group (NIPG)
ELTE-IK, Department of Artificial Intelligence

- Worked on **video processing and object tracking** as part of project EFOP-3.6.3-VEKOP-2017-00002, resulting in two rank A conference publications at IJCNN.
- Developed an **AI-based eye blink detection** pipeline for **non-verbal communication analysis**, published in a Q2 journal, and later integrated into a composite AI framework for **automatic behavior analysis** of children with autism spectrum disorder, resulting in a subsequent Q1 journal publication.
- Partnered with *Barcelona University* on **multimodal sentiment and personality perception** research, leading to publication and strengthening NIPG partnerships.
- Gained **multimodal data collection** experience through collaboration with the *ELTE Faculty of Education and Psychology* on a Still Face experiment **analyzing infant–mother communication**, and contributing to 2 BSc and 2 MSc theses.
- Developed expertise in **speech processing** through my MSc thesis research on **speech de-identification**, which was recognized with TDK awards and a publication.
- **Teaching MSc courses** on the mathematical foundations and applications of state-of-the-art AI (Large Language Models included), **supervising** students (BSc and MSc), and guiding a MSc student's TDK project.
- **Presented seminars** at prestigious venues, including the *John von Neumann Computer Society (NJSZT)*, leading Hungarian universities such as the *University of Szeged*, the *CNC seminar at Comenius University* in Bratislava, as well as the renowned *K&H Bank*.
- Regularly contributed as a **poster presenter** at prominent Hungarian events, including *Hungarian Machine Learning Days* and *MILAB Workshops*, enhancing the visibility and reputation of NIPG and ELTE.
- Managed 40+ research databases supporting 10+ NIPG projects over the past 5 years.

Associate Scientist — 07/2014 – 08/2017

Institute for Computer Science and Control
Hungarian Academy of Sciences

- **Executed a basic research project** for *OPEL Szentgotthárd Kft.* as part of iKOMP, applying feature selection and production trend forecasting methods to reduce engine failures.
- **Developed a novel feature selection technique** called AHFS, published in a Q1 journal, and successfully **applied in real-world tasks** such as situation detection in specialized machining and wind turbine bearing temperature monitoring.
- **Designed a software framework** for AI model training and data visualization as part of my BSc thesis, extending prior tools to support data mining; created in *collaboration with MTA SZTAKI* and adopted as a successor to their in-house system.

Selected Publications

- Bruno Carlos Dos Santos Melício, Kaan Karaköse, **Ádám Fodor**, Linyun Xiang, Viktor Varga, Latha Soorya, Emily Dillon, Péter Kun, András Sárkány, Mohamed Chetouani, Kristian Fenech; *Multimodal Framework for Automatic Behavior Analysis of Children with Autism During ADOS-2, Cognitive Computation*, vol. 17, 137, 2025
- **Ádám Fodor**, Kristian Fenech, András Lőrincz; *BlinkLinMulT: Transformer-Based Eye Blink Detection*, MDPI Journal of Imaging, vol. 9-10, 196, 2023
- Zsolt János Viharos, Krisztián Balázs Kis, **Ádám Fodor**, Máté István Büki; *Adaptive, hybrid feature selection (AHFS)*, Pattern Recognition, vol. 116, 107932, 2021
- [For further 7 publications, please visit my website.]

Education

PhD candidate in Computer Science — 08/2019 – 08/2025

Eötvös Loránd University (ELTE)

- Thesis: Affective Behavior and Social Interaction Analysis
- Supervisor: Dr. habil. András Lőrincz
- Submitted, successful departmental defense, public defense is upcoming

MSc in Computer Science — 01/2016 – 07/2018

Eötvös Loránd University (ELTE)

- Thesis: Speech de-identification with deep neural networks
- Supervisor: Dr. habil. András Lőrincz
- Qualification: excellent

References

Dr. Kristian Fenech
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Dr. habil. András Lőrincz
Founder of NIPG, ELTE
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